Jay A. Panchal

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Summary

Software Engineer finishing a Master of Science (MS) in Computer Science in December 2026, with 1.5+ years building production backend systems, REST APIs, and full-stack applications in Java, Python, and JavaScript. Builds AI systems with Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and PyTorch. AWS Certified, and eager to grow on a strong engineering team.

Education

Loyola Marymount University — Master of Science in Computer Science *Los Angeles, CA | Expected Dec 2026*
GPA: 3.88 | *Coursework*: Generative AI, Machine Learning, Database Systems, Deep Learning, Algorithms

Professional Experience

Software Engineer Intern — Date Maroon *Los Angeles, CA (Remote) | Aug 2026 – Present*
- Ship production features end to end for a startup's mobile and web apps (onboarding, profiles, assessments, matching), from implementation through code review and release
- Build React Native screens for iOS and Android on top of backend REST APIs, and review React and Next.js code before each merge
- Debug failures in core user flows from server logs and monitoring dashboards to find the root cause
- Use Claude Code and GitHub Copilot daily to build features from specs and acceptance criteria, working closely with founders, product, and design

Teaching Assistant (Machine Learning & Deep Learning) — Loyola Marymount University *Los Angeles, CA | Aug 2025 – Present*
- Mentor students debugging Python machine learning code in lab sessions and grade assignments with clear written feedback

Software Engineer — Aequitas Infotech Pvt. Ltd. *Ahmedabad, India | Jan 2024 – Aug 2024*
- Engineered production backend features with Java Spring Boot and MySQL for a practice management platform serving tens of thousands of ICAI-certified Chartered Accountants
- Designed and optimized relational SQL schemas, improving the reliability of complex cross-table lookups
- Built Selenium, Postman, and Appium test suites and maintained Jenkins CI/CD pipelines for build and deployment
- Resolved client-reported production issues through pair programming, and trained end users in sessions of up to 300 people

Backend Developer — CodeHabit (Startup) *Gandhinagar, India | Sep 2023 – Dec 2023*
- Built REST APIs with Node.js and Express for a hospital management system serving 25+ staff, secured with JWT and role-based access control
- Designed MongoDB and MySQL schemas with pagination and versioning, working with frontend developers on API contracts

Software Engineer Intern — Praxware *Ahmedabad, India | Jan 2023 – Aug 2023*
- Developed Django and MySQL backend workflows for a sales and inventory system, plus Power BI dashboards for stakeholders

Projects

GridWatch — Energy Affordability Intelligence System | Python, XGBoost, SHAP, RAG, FAISS, OpenAI API, Streamlit Live Demo
- Built an end-to-end RAG pipeline with a FAISS vector database over government policy documents, grounding LLM answers in retrieved sources
- Trained XGBoost on Census, EIA, and NOAA data to predict energy-burden risk across 50 California cities, with SHAP explanations, and shipped it as a Streamlit app with an AI research assistant that cites sources

GoTinerary — AI Trip Planner | Gemini API, Google Search Grounding, Google Places API |**LA Hacks 2026 Best Use of Gemini (MLH)**
- Built solo in one weekend an agentic AI application where Gemini calls the Google Places API through tool calling to re-plan the day
- Grounded every suggestion in verified Places data with a scheduling engine for real opening hours; won Best Use of Gemini API at LA Hacks 2026

atlas.db — Database Benchmarking System | PostgreSQL, MongoDB Atlas, Amazon RDS, Python, PostGIS, Power BI
- Benchmarked PostgreSQL against MongoDB on 622K Los Angeles traffic records, writing the Python ingestion scripts; PostgreSQL won 4 of 5 analytical queries (1.72s vs 92.64s)
- Won Most Effective Presentation Delivery and Most Cohesive System Architecture awards

Technical Skills

- **Languages**: Python, Java, TypeScript, JavaScript, Go, C, C++, SQL
- **Frameworks/Libraries**: Node.js, Express, Spring Boot, Django, Flask, FastAPI, React Native, Angular, PyTorch, scikit-learn, LangChain
- **AI/LLMs**: LLMs, RAG, FAISS, Vector Databases, OpenAI API, Anthropic Claude API, Google Gemini API, Tool Calling, Model Context Protocol (MCP), LoRA Fine-Tuning
- **Databases/Cloud**: PostgreSQL, MySQL, MongoDB, AWS (RDS, S3, Lambda), Docker, Linux
- **Tools**: Git, GitHub Actions, Jenkins, CI/CD, Selenium, Postman, Appium, Jira, Claude Code, Cursor, GitHub Copilot
- **Certification**: AWS Certified Cloud Practitioner

Awards

- **LA Hacks 2026 — Best Use of Gemini API (MLH)** — *Oct 2026*: Built GoTinerary solo in one weekend, an agentic AI trip planner using Gemini tool calling and the Google Places API
- **LMU Datathon (Southern California)** — *3rd Place, (March 2026)*: Analyzed the NLSY97 dataset using pandas, regression, and statistical tests to wage gap insights, placing 3rd among competing universities
- **LMU BEST Entrepreneurship Bootcamp** — *3rd Place, (Jan 2026)*: Built GOTINERARY, an AI-powered dynamic travel itinerary app, applying business model canvas and customer segmentation across a 4-person team against 7 competing teams